

## Overview

The **IMX-6™** is a 10-DOF sensor module consisting of a tactical-grade Inertial Measurement Unit (IMU), magnetometer, and barometer. Output includes angular rate, linear acceleration, magnetic vector, and barometric pressure and altitude. IMU calibration consists of bias, scale factor, cross-axis alignment, and temperature compensation. The IMX-6 includes Attitude Heading Reference System (**AHRS**) sensor fusion to estimate roll, pitch, and heading. Adding GNSS input to the IMX-6 enables onboard Inertial Navigation System (**INS**) sensor fusion for roll, pitch, heading, velocity, and position.

The **RUG-4-IMX-6™** series adds a rugged aluminum enclosure and RS232, RS485, and CAN bus to the IMX-6.

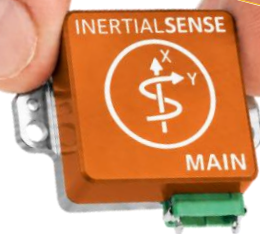
The **RUG-4-IMX-6-RTK™** includes a multi-frequency GNSS receiver with RTK precision position enabling INS sensor fusion for roll, pitch, heading, velocity, and position.

The **RUG-4-IMX-6-Dual™** includes two multi-frequency GNSS receivers with RTK precision position and dual GNSS heading/compass.

The **Inertial Sense SDK** is an open-source software development kit for quick integration to configure and communicate with Inertial Sense products. The SDK includes data logger, math libraries, and interface for Linux, Windows, and embedded platforms.

## Applications

- Drone Navigation
- Unmanned Vehicle Payloads
- Ground and Aerial Survey
- Automotive Navigation
- Stabilized Platforms
- Antenna and Camera Pointing
- First Responder and Trackers
- Health and Fitness Monitoring
- Robotics and Ground Vehicles
- Maritime



**RUG-4-IMX-6**  
 Size: 30.5 x 25.4 x 9.9 mm  
 Weight: 10.6 g



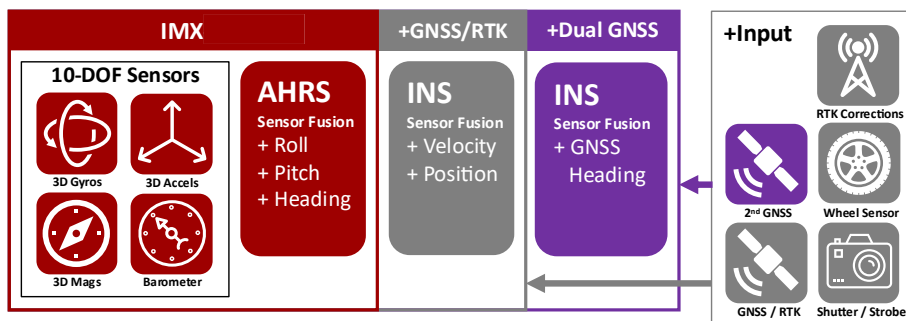
**IMX-6**  
 Size: 15.6 x 12.5 x 2.9 mm  
 Weight: 0.9 g  
 INS: External GNSS Input



**RUG-4-IMX-6-RTK/Dual**  
 Size: 30.5 x 25.4 x 14.8 mm  
 Weight: 14 g  
 GNSS: All-Band L1/L2/L5/L6

## Features

- **Tactical-Grade IMU**
  - Gyro: 1.5 °/hr Bias Instability, 0.10 °/vhr ARW
  - Accel: 2.8 µg Bias Instability, 0.013 m/s/vhr VRW
- **0.03° Dynamic Roll/Pitch**
- **0.09° Dynamic Heading**
- **IMU Sensing Range: ±4000 °/s, ±16/80/320 g**
- **Reflow-Compatible PCB Module**
- **Output Data Rates: 1000 Hz IMU, 500 Hz INS**
- **30% More Accurate Navigation Than IMX-5**
- Pin-compatible with IMX-5
- External GNSS Input Support (Multi-Band)
- Attitude (Roll, Pitch, Yaw, Quaternions), Velocity, and Position Synchronized to UTC
- 5 Redundant IMUs Calibrated for Bias, Scale Factor, Cross-axis Alignment, and G-Sensitivity
- IMU Shock and Fault Rejection
- -40°C to 85°C Sensor Temperature Calibration
- Binary and NMEA ASCII Protocol
- Barometric Pressure and Humidity
- Strobe In/Out Data Sync (Camera Shutter Event)
- Fast Integration with SDK and Example Software
- RUG-4-IMX-6: RS232, RS485, CAN bus



## Specifications

| Performance (AHRS, INS, RUG-4)       | Typ   |                                     |
|--------------------------------------|-------|-------------------------------------|
| INS Dynamic Roll/Pitch** (RMS)       | 0.03° | (GNSS-aided with motion excitation) |
| Static Roll/Pitch (RMS)              | 0.09° |                                     |
| INS Dynamic Heading** (RMS)          | 0.09° | (GNSS velocity-derived)             |
| Static Heading w/Dual Compass* (RMS) | 0.4°  |                                     |
| Static Heading w/magnetometer (RMS)  | 1.0°  |                                     |

\*1 m baseline distance between GNSS antennas.

\*\*With GNSS input and periodic motion >0.8 m/s<sup>2</sup> acceleration and >2 m/s velocity.

| Performance (INS, RUG-4)          | RUG-4     | +RTK             |
|-----------------------------------|-----------|------------------|
| Horizontal Position (w/ SBAS)     | 1.5 m CEP | 1 cm + 1 PPM CEP |
| Velocity (GPS and INS)            | 0.02 m/s  |                  |
| Angular Resolution                | 0.05°     |                  |
| Operation Limits                  |           |                  |
| Velocity (external GNSS)          | 500 m/s   |                  |
| Altitude (external GNSS)          | 50 Km     |                  |
| Altitude (Barometric)             | 10 Km     |                  |
| GNSS cold-start time to first fix | 24 s      | -                |

| Performance                               | Typ          |
|-------------------------------------------|--------------|
| Startup Time                              | 0.8 s        |
| INS/AHRS Timestamp Accuracy (RMS)         | 1 us         |
| Max Output Data Rate (IMU, AHRS/GNSS-INS) | 1000, 500 Hz |
| IMU signal latency @ 1 KHz ODR            | 4 ms         |

| Absolute Maximum Ratings        | MAX                        |
|---------------------------------|----------------------------|
| Acceleration                    | 10,000 g                   |
| Ambient Operating Temperature   | -40 to 85 °C               |
| Junction Temperature            | -40 to 105 °C              |
| Storage Temperature             | -40 to 125 °C              |
| Overpressure                    | 600 kPa                    |
| ESD rating (Human body model)   | ± 2 kV                     |
| Solder Reflow Temperature Max   | 245 °C                     |
| Solder Reflow Temperature Limit | 217 °C liquidus: 40 – 60 s |

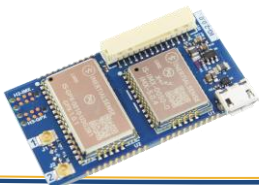
| IMX-6.0 Sensors             | IMU - Gyros   | IMU - Accels   | Mags      | Pressure    |
|-----------------------------|---------------|----------------|-----------|-------------|
| Operating Range             | ±4000 °/s     | ±16/80/320* g  | ±2500 μT  | 30–125 kPa  |
| In-Run Bias Stability**     | < 1.5 °/hr    | < 2.8 μg       |           |             |
| Random Walk: ARW, VRW       | 0.10 °/√hr    | 0.013 m/s/√hr  |           |             |
| Non-linearity               | 0.015 % FSR   | 0.015 % FSR    |           |             |
| Noise Density               | 3.8 mdps/√Hz  | 46 μg/√Hz      | 20 nT/√Hz | 0.18 Pa/√Hz |
| Bias Error over -40C to 85C | 0.3 °/s RMS   | 3.7 mg RMS     |           |             |
| Max Output Rate             | 1 KHz         | 1 KHz          | 50 Hz     | 50 Hz       |
| Bandwidth                   | 539 Hz        | 416 Hz         | 50 Hz     | 5 Hz        |
| Alignment Error             | 0.03°         | 0.03°          | 0.05°     |             |
| Resonant Frequencies        | 19-29 KHz     | 4,11,15-24 KHz |           |             |
| Sampling Rate               | 8 KHz         | 4 KHz          | 300 Hz    | 200 Hz      |
| Resolution                  | ***0.0076 °/s | ***122 μg      | 0.3 μT    | 0.03 Pa     |

\*Contact sales for extended sensing range

\*\*Performance specifications per IEEE-STD-952-1997

\*\*\*1 KHz resolution after oversampling

| Function                          | IMX™ | +RTK | +Dual |
|-----------------------------------|------|------|-------|
| Gyro & Accelerometer (IMU)        | •    | •    | •     |
| Magnetometer & Barometer          | •    | •    | •     |
| Roll, Pitch, Heading (AHRS)       | •    | •    | •     |
| Heading, Velocity, Position (INS) |      | •    | •     |
| GNSS Heading                      |      |      | •     |



**Development kits  
available on our  
website.**

| Electrical (IMX-6)             | Min  | Typ | Max | Units |
|--------------------------------|------|-----|-----|-------|
| Power Draw                     |      |     |     |       |
| IMU @ 1 KHz                    |      | 105 | 115 | mW    |
| w/ AHRS, INS @ 250 Hz          |      | 110 | 120 | mW    |
| Supply Voltage (Vcc)           | 3.0  | 3.3 | 3.6 | V     |
| I/O Pin MAX Voltage Range      | -0.5 |     | 3.6 | V     |
| Total Output Current, All Pins |      |     | 100 | mA    |
| I/O Pin Output Current         |      |     | 20  | mA    |
| I/O Pin Input low-level        | 0.99 |     |     | V     |
| I/O Pin Input high-level       | 2.31 | 3.3 | 3.6 | V     |
| I/O Pin Output high-level      |      | 3.3 |     | V     |
| STROBE input frequency         |      |     | 1   | KHz   |
| STROBE output jitter           |      | 10  |     | us    |
| Rising Slope of VIN*           | 2.4  |     |     | V/ms  |

\*The supply rising slope must be higher than minimum rating for proper function.

| Electrical (RUG-4)        | Min | Typ  | Max | Units |
|---------------------------|-----|------|-----|-------|
| Supply Voltage (VIN)      | 3.1 |      | 23  | V     |
| With R1 SMT Jumper Loaded |     |      | 42  | V     |
| RUG-4-IMX-6-RTK + Antenna |     |      |     |       |
| Current Draw @ 5V, 250Hz* |     | 195  |     | mA    |
| Power Consumption @250Hz* |     | 937  |     | mW    |
| Power Consumption @100Hz* |     |      |     | mW    |
| Power Consumption – Dual  |     | 1480 |     | mW    |

\*Navigation filter update rate.

| Mechanical (IMX-6)                                                        | Typ               | Units |
|---------------------------------------------------------------------------|-------------------|-------|
| Size                                                                      | 15.6 x 12.5 x 2.9 | mm    |
| Weight                                                                    | 0.9               | grams |
| Effective Thermal Resistance (Θ <sub>eff</sub> )                          | 15                | °C/W  |
| Thermal Resistance (Θ <sub>JA</sub> , Θ <sub>JB</sub> , Θ <sub>JC</sub> ) | 42.3, 27.5, 1.6   | °C/W  |

| Mechanical (RUG-4) | Units                                        | Conditions                    |
|--------------------|----------------------------------------------|-------------------------------|
| Size               | 30.5 x 25.4 x 9.9<br>30.5 x 25.4 x 14.8      | mm<br>RUG-4<br>RUG-4-RTK/Dual |
| IP Rating          | 40                                           | No liquid protection          |
| Mounting Tab       | 30.836                                       | mm                            |
| Hole Spacing       |                                              |                               |
| Weight             | 14.0                                         | grams                         |
| Connectors         | Main: Harwin# G125-MV11205L1P, GPS 1/2: MMCX |                               |

| Communications & I/O    |                                       |
|-------------------------|---------------------------------------|
| IMX-6 Interface         | USB, UART x3, SPI                     |
| RUG-4 Interface         | USB, UART x2, RS232, RS485, CAN*, SPI |
| Max Baud Rate:          |                                       |
| SPI                     | 10 Mbps                               |
| UART, RS422, RS485      | 10 Mbps                               |
| RS232                   | 500 Kbps                              |
| Strobe Inputs / Outputs | 4 / 1                                 |

\* Available in future firmware release.

